

EMCH 201: Introduction to Application of Numerical Methods

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Assignment 1

Assignment due date: **11:59 PM, Monday September 6th, 2021**

Point Possible: 100 (20+30+40+20)

Submission requirements:

- Detailed problem-solving process needs to be provided for full points
- Upload readable electronic copy to blackboard in a single file
- Late submission will not be graded

- Please specify the significant figures for the following number (10pts):
 - 0.1210
 - 100.0E+3
 - 2.12
 - 0.212
 - 10000.0
- Based on Taylor Series, please prove $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ (Hint: evaluate $\sin(x)$ at the expansion point of 0) (30 pts)
- Use a forward difference and a centered difference approximation to estimate the first derivative of the function: $f(x) = 10x^4 - 4x^3 + 3x^2 + 6x$. Evaluate the derivative at $x = 1$ using a step size of $h = 0.1$ (that is, $x_{i+1} = 1.1$, $x_i = 1$, and $x_{i-1} = 0.9$). Compare your results with the true value of the derivative and compute the true percent relative error ε_t . (40 pts)
- Evaluate and interpret the condition numbers for
 - $f(x) = e^{-x^2} + x^2$ for $x = 10$ (10 pts)
 - $f(x) = \frac{\sqrt{x^3+1}}{4} - 2x$ for $x = 300$ (10 pts)